

Review

Adherence to and Retention in Medications for Opioid Use Disorder Among Adolescents and Young Adults

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The volatile opioid epidemic is associated with higher levels of opioid use disorder (OUD) and negative health outcomes in adolescents and young adults. Medications for opioid use disorder (MOUD) demonstrate the best evidence for treating OUD. Adherence to and retention in MOUD, defined as continuous engagement in treatment, among adolescents and young adults, however, is incompletely understood. We examined the state of the literature regarding the association of age with adherence to and retention in MOUD using methadone, buprenorphine, or naltrexone among persons aged 10–24 years, along with related facilitators and barriers. All studies of MOUD were searched for that examined adherence, retention, or related concepts as an outcome variable and included adolescents or young adults. Search criteria generated 10,229 records; after removing duplicates and screening titles and abstracts, 587 studies were identified for full-text review. Ultimately, 52 articles met inclusion criteria for abstraction and 17 were selected for qualitative coding and analysis. Younger age was consistently associated with shorter retention, although the overall quality of included studies was low. Several factors at the individual, interpersonal, and institutional levels, such as concurrent substance use, MOUD adherence, family conflict, and MOUD dosage and flexibility, appeared to have roles in MOUD retention among adolescents and young adults. Ways MOUD providers can tailor treatment to increase retention of adolescents and young adults are highlighted, as is the need for more research explaining MOUD adherence and retention disparities in this age group.

adolescents; buprenorphine; medications for opioid use disorder; methadone; naltrexone; opioid use disorder; retention; young adults

Abbreviations: LMIC, low- and middle-income countries; MOUD, medications for opioid use disorder; OUD, opioid use disorder.

INTRODUCTION

Opioid misuse has quickly become a major public health concern globally. Most high-income countries already are experiencing epidemic rates of opioid misuse, and rates are increasing in low- and middle-income countries (LMIC) (1). According to the most recent Global Burden of Disease Study (2), opioid dependence is estimated to be the most common substance use disorder worldwide after alcohol use disorder and poses a considerable burden in the form of disability-adjusted life-years, responsible for 86,200 deaths and 3,656,100 years of life lost.

Opioid misuse is increasing among young people between the ages of 10 and 24 years, especially in high-income

countries, mainly due to increasing availability and non-medical use of prescription opioids (3, 4). This population is especially vulnerable to negative consequences of opioid misuse, developing dependence faster than adults as a result of being in a life period critical for cognitive, emotional, and social development (5).

Medications for opioid use disorder (MOUD) are the most effective treatment for chronic opioid use disorder (OUD) and includes pharmacotherapies like methadone, buprenorphine, and naltrexone, all of which work on the μ opioid receptor (6). Both methadone and buprenorphine act as opioid agonists, activating the μ opioid receptor and alleviating opioid withdrawal. Unlike methadone, which is a

full agonist, buprenorphine is a partial agonist, which means it has a ceiling effect, limiting it from producing the same effects as a full agonist and resulting in a lower likelihood of producing euphoria and of causing opioid overdose and other negative effects associated with opioid consumption. Naltrexone, on the other hand, is an opioid antagonist, which means it blocks the μ opioid receptor, preventing opioids from taking effect if used. As such, rather than alleviate opioid withdrawal, naltrexone functions to prevent relapse (7).

In industrialized and developing countries, MOUD consistently curtail the negative health, social, and legal consequences of OUD (8, 9). More specifically, MOUD reduce opioid use and subsequent risk of HIV and hepatitis C transmission, while improving mental health and reducing drug-related and all-cause mortality (3, 9–14). Despite their remarkable success, MOUD remain either absent or inadequately scaled to need in most countries (1, 15, 16). Despite recommendations of optimal MOUD coverage of 40%, with minimal effective coverage at 20%, as few as an estimated 16 persons per 100 persons who inject drugs receive MOUD globally, with high variability observed (1, 16).

The evidence base demonstrating the effectiveness of MOUD among adolescents and young adults is limited and mostly focuses on buprenorphine (17); that said, multiple international agencies, professional groups, and other institutions, such as the World Health Organization and the American Academy of Pediatrics (18, 19), recommend the use of MOUD for individuals in this age group. Despite these recommendations, many young people with OUD still lack access to MOUD (20–22). Even when they are able to access MOUD, young people still experience challenges with adherence to and retention in MOUD (23, 24). As such, there is an urgent need to identify the factors causing young people to prematurely drop out of MOUD therapy.

In previous systematic reviews, authors have explored the factors associated with MOUD initiation and outcome among young persons (25, 26) or have explored barriers to retention in MOUD—defined as continuous engagement in treatment—among the general population (27, 28). None to our knowledge have examined specifically which factors are responsible for adherence to and retention in MOUD among young persons. In addition, a global perspective on this issue, which could highlight different approaches and their effectiveness depending on the social context, development status, and policies and procedures related to MOUD, is also lacking. Here, we review the research on the influence that age has on adherence to and retention in MOUD among adolescents and young adults globally.

METHODS

We conducted a systematic review of all studies of MOUD including evidence of adherence, retention, or related concepts (e.g., compliance, dropout, attrition) as an outcome variable and in which individuals between the ages of 10 and 24 years were part of the study sample. This age range was chosen because it is consistent with the definition of young people used by the World Health Organization and

other international agencies (4). Adherence was defined primarily by continued consumption of MOUD. Adherence (also called compliance) is sometimes measured through attendance, especially in methadone programs where consumption is observed. It also may be confirmed through urinalysis, which can test for metabolites of MOUD. Retention was defined by uninterrupted engagement in treatment, operationalized in a variety of ways, such as by attendance at program appointments or renewal of prescriptions. According to this definition, retention could be defined as specific time points (e.g., at 3 months or 1 year) or as the time from treatment initiation to dropout. As may be evident from the latter measurement, retention in MOUD is also sometimes defined by its opposites: treatment attrition or dropout. These definitions were used to broadly capture the full scope of studies examining these concepts as treatment outcomes. Standard definitions for these terms in the context of MOUD were not found in other systematic reviews on the topic.

The methods used for this review are largely summarized in the International Prospective Register of Systematic Reviews protocol (29). A search strategy was tested against a set of validation articles, then used in 7 bibliographic databases: CINAHL, Criminal Justice Abstracts, Embase, Medline via Ovid, PsycINFO, Scopus, and Web of Science Core Collection. Similar search strategies were used in each database, adjusted for the subject headings available in each database's controlled vocabulary schema. The searches took place at between December 20, 2018, and January 3, 2019.

We searched for articles using title-abstract keywords and, where available, controlled vocabulary for 4 concepts: the population of adolescents and young adults, OUD, MOUD, and adherence or retention (Web Appendix) (available at <https://academic.oup.com/epirev>). In databases with subject indexing, animal studies that did not involve humans were excluded. No language or date restriction was applied.

After removing as many duplicate articles as possible, title/abstract review was conducted to identify potentially relevant studies. Inclusion criteria required articles to focus on MOUD; include data on adherence, retention, or related concepts as an outcome among individuals aged 10 to 24 years (or people identified by the authors as adolescents or young adults); be an original, peer-reviewed research article; provide original data (i.e., commentaries, editorials, letters, opinions, and reviews were excluded); and have been published in a peer-reviewed journal. No publication date restriction was applied and only studies published in Bosnian, Croatian, English, Italian, Montenegrin, Russian, Serbian, or Spanish were included in the final review. These languages were selected on the basis of the fluency of the members of the research team, which viewed including articles in as many languages as possible so important research studies could be included from non-English-speaking countries, given the international focus of this review. Covidence (Melbourne, Victoria, Australia) was used for both title and abstract and full-text review. Disagreements around inclusion of articles in the title and abstract or full-text review was resolved by a third member of the research team.

The research team systematically searched the gray literature, such as relevant government websites (e.g., National Institute on Drug Abuse, Substance Abuse Mental

Health Services Administration, United Nations Office of Drugs and Crime), and provider organizations (e.g., American Academy of Pediatrics, American Society of Addiction Medicine) for any documents or data they may provide. Google Scholar was used to identify validation articles to ensure quality of the review.

From all relevant sources, the research team extracted data related to adherence to or retention in MOUD among young persons, as defined at the beginning of Methods, or data on the association between age and these outcomes. Information about the research study and covariates was also collected.

Two members of the research team were required to agree on study inclusion during both title and abstract review and full-text review. Members of the research team individually

extracted data, which were double-checked by another member of the research team to ensure quality of data extraction. Two members of the research team conducted qualitative coding with a subset of the final articles that focused solely on adolescents and young adults; this coding was reviewed and analyzed by a third member of the research team, who developed themes on the basis of an adapted version of the Social Ecological Model (30), a framework for organizing into nested levels factors that affect behavior. This framework was selected because of the way it represents the relationships among the individual, interpersonal, institutional, and structural factors that may influence adherence to and retention in MOUD among adolescents and young adults. Quality assessment was conducted independently by a pair of researchers using the Grading of Recommen-

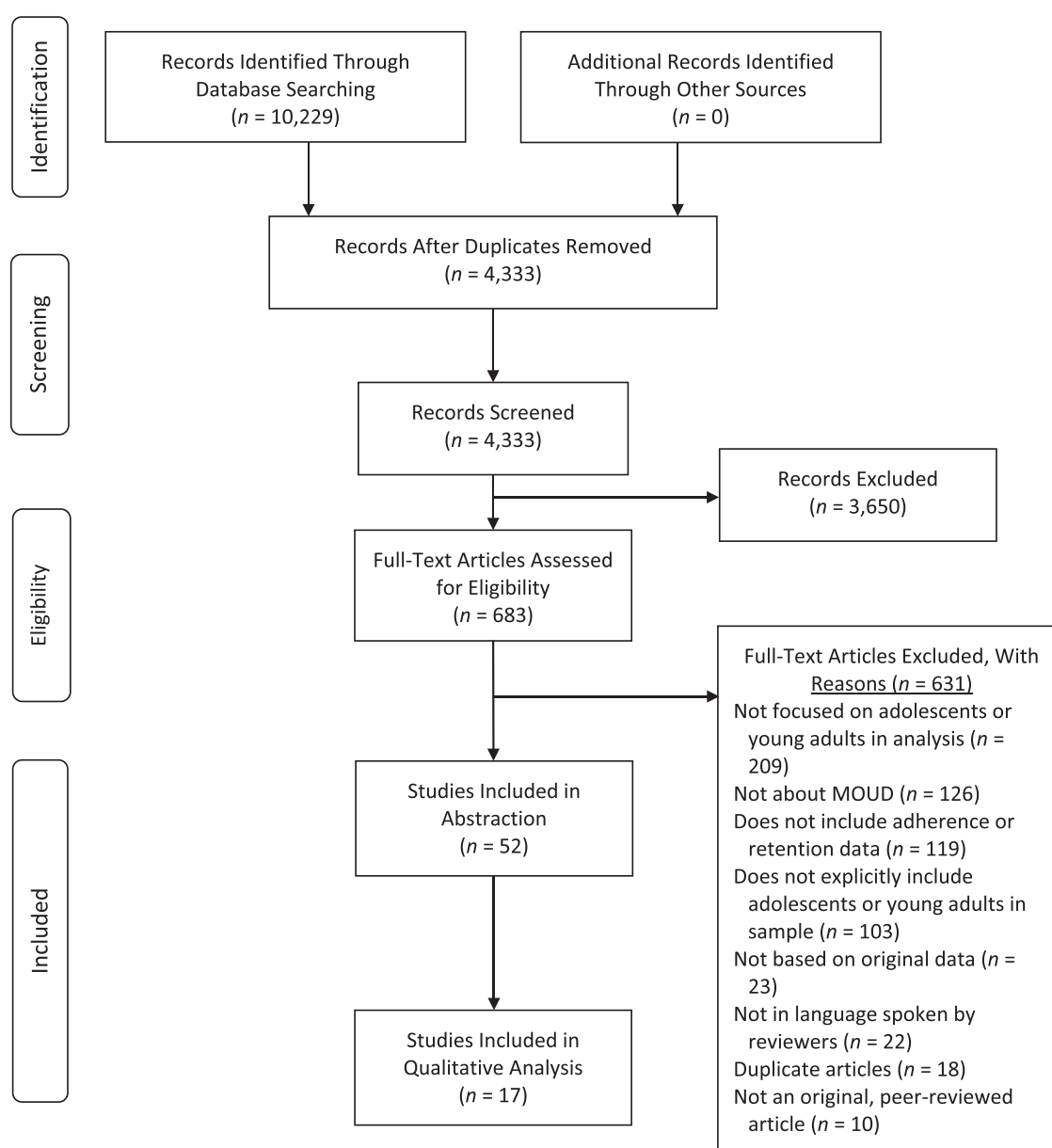


Figure 1. Preferred Reporting Items for Systematic Reviews and Meta-Analyses flow chart. MOUD, medications for opioid use disorder.

dations Assessment, Development, and Evaluation system and guidelines (31, 32). Citation analysis was conducted on April 8, 2019, on this final subset of 17 articles using Web of Science Core Collection to identify articles that cited those included articles. After deduplication of that set, 181 citations remained. Those citations were processed through title and abstract review and full-text screening. None of the studies identified met inclusion criteria and so no additional articles were added to the synthesis.

RESULTS

The initial search yielded 10,229 articles, of which 5,896 were immediately identified as duplicates. The title and abstracts of a total of 4,333 articles were screened against inclusion criteria, leaving 683 articles for full-text review. A total of 52 articles were included in qualitative synthesis after full-text review because the studies they reported included adolescents or young adults in the sample and examined the relationship between age and retention either continuously or in a categorical way in which the results specific to this age group could be examined. Seventeen studies were focused on adolescents and/or young adults specifically; the results from these studies were coded and analyzed qualitatively. Figure 1 presents the flow of studies through the review according to Preferred Reporting Items for Systematic Reviews and Meta-Analyses guidelines (33).

More than half of the included studies had been conducted in the United States ($n = 30$). The majority of the other studies were conducted in other high-income countries, including Australia ($n = 4$), Ireland ($n = 3$), Canada ($n = 2$), France ($n = 2$), Israel ($n = 1$), Italy ($n = 1$), Norway ($n = 1$), and Switzerland ($n = 1$). Only 7 studies were conducted in low- and middle-income countries (LMIC), including 3 studies in India and 1 each in Afghanistan, Tanzania, Turkey, and Ukraine.

More than a third of the included studies assessed adherence to or retention in buprenorphine only ($n = 21$) or methadone only ($n = 16$). Four studies assessed retention in treatment with naltrexone only, 2 with oral naltrexone, and 2 with injectable naltrexone. Some studies assessed both methadone and buprenorphine ($n = 8$), buprenorphine and injectable naltrexone ($n = 1$), methadone and oral naltrexone ($n = 1$), or all 3 together ($n = 1$).

No other studies were identified through additional gray literature searches. Information about the 52 articles included in qualitative synthesis is reported in Table 1.

Relationship between age and retention

Most of the 35 studies ($n = 20$; 57.1%) that examined age as a predictor of retention found that adolescents and young adults are at increased risk of dropout from MOUD treatment. Chaudhry et al. (34) did initially find younger age was associated with increased retention in MOUD; however, multivariate analysis determined this association was due to confounding variables. Cousins et al. (35) found younger age was associated with longer retention in MOUD even after adjusting for other variables, with individuals aged 40–

65 years having 25% greater odds of discontinuing MOUD compared with individuals aged 16–19 years. Almost one-third of studies ($n = 11$, 31.4%) found age was not a statistically significant predictor of adherence or retention in MOUD among adolescents and young adults (34, 36–45). All these studies either were conducted in LMIC or had a small sample size (largest sample size outside of studies done in LMIC, $n = 981$) (36).

Quality assessment

The quality of the articles exploring the association between age and retention in MOUD was low according to the guidelines outlined in the Grading of Recommendations Assessment, Development, and Evaluation system (32). This was because few of the studies in included articles were randomized controlled trials. Overall, the researchers who conducted quality assessment agreed that the studies, as a group, demonstrated low risk of overall bias and no evidence of publication bias. Indirectness was not seen as an issue in these studies. A moderate degree of inconsistency was observed across study findings, though it is notable that the studies in which no evidence was found of association between age and retention in MOUD were of lower quality and exhibited a greater degree of imprecision than the studies that observed a positive association between age and retention in MOUD, due largely to differences in sample sizes between the 2 sets of articles. Studies in which older age was associated with longer retention ($n = 20$) averaged a sample size of almost 5,000 participants, whereas the studies in which no significant association was observed between age and retention ($n = 11$) had an average sample size more than 7 times smaller, at 709 participants. As such, it is possible that some of the studies in which no association was found between age and retention did not have sufficient sample size to identify significant results.

In only 1 study was lower age correlated with increasing retention (35). Of the remaining 3 articles, 1 was focused on adherence as an outcome (46), 1 reported mixed results by sex (47), and the last reported mixed results on the basis of the time frame, with a positive association identified between age and retention at 6 months that was no longer significant at 12 months (48). None of these studies were included in either of the subgroups in quality assessment. The results of quality assessment are presented in Table 2.

Relationship between age and MOUD retention by MOUD type

To explore the relationship between age and retention in MOUD, we stratified studies by MOUD type (i.e., methadone, buprenorphine, naltrexone). After excluding the studies with mixed findings, described previously, we found a total of 20 studies had examined this relationship for methadone, 15 for buprenorphine, and 2 for naltrexone. The sum of the studies listed here is greater than the total number of studies examining the relationship between age

Table 1. Final Articles Analyzing Adherence and/or Retention, Stratified by Age

First Author, Year (Reference No.)	Country	No. of Participants	Study Design	Type of MOUD	Max. Retention Time Frame	Age-Related Findings
Anderson, 2004 (62)	Canada	5,087	Program evaluation	Methadone only	12 months	Older age was associated with longer retention.
Armstrong, 2010 (42)	India	2,569	Operational research	Buprenorphine only	1 year	Age was not significantly associated with retention.
Babst, 1971 (47)	United States	679	Secondary analysis of program data	Methadone only	2 years	Older age was associated with longer retention in men and with shorter retention in women.
Bartu, 2002 (36)	Australia	981	Retrospective cohort study	Oral naltrexone only	87 weeks	Age was not significantly associated with retention.
Bell, 2006 (60)	Australia	61	File review	Methadone, buprenorphine	185 days	All participants in the study were adolescents. Methadone maintenance was more effective than buprenorphine at retaining adolescents in treatment.
Burns, 2009 (63)	Australia	42,690	Retrospective cohort study	Methadone, buprenorphine	9 months ^a ; 12 months	Older age was associated with longer retention and had the largest association with retention.
Burns, 2015 (64)	Australia	15,600	Retrospective cohort study	Methadone, buprenorphine	12 months	Older age was associated with longer retention.
Chaudhry, 2012 (34)	United States	142	Retrospective cohort study	Oral naltrexone only	≥ 17 weeks ^a ; 53 to ≥ 72 weeks	Older age was associated with shorter retention, but this relationship did not remain so after adjusting for potential confounders.
Condelli, 1993 (65)	United States	526	Secondary analysis of prospective study data	Methadone only	1 year	Older age was associated with longer retention and was the only significant predictor.
Cousins, 2017 (35)	Ireland	6,393	Retrospective cohort study	Methadone only	≥ 1 year	Older age was associated with shorter retention.
Damian, 2017 (66)	United States	445	Retrospective cohort study	Buprenorphine only	90 days	Older age was associated with longer retention.
Dayal, 2017 (57)	India	68	Retrospective cohort study	Buprenorphine only	1 year	All participants in the study were young adults. Close family member substance use and buprenorphine dose were associated with shorter retention, whereas recent history of injection drug use was positively associated with longer retention.
Deck, 2005 (67)	United States	5,308	Retrospective cohort study	Methadone only	1 year	Older age was associated with longer retention.
del Rio, 1997 (43)	Switzerland	111	Prospective cohort study	Methadone only	54 months	Older age was associated with longer retention but only at the univariate level. Age was not significantly associated with retention in multivariate analysis.

Table continues

Table 1. Continued

First Author, Year (Reference No.)	Country	No. of Participants	Study Design	Type of MOUD	Max. Retention Time Frame	Age-Related Findings
Dhawan, 2013 (37)	India	231	Intervention study	Buprenorphine only	9 months ^a	Age was not significantly associated with retention.
D'Ippoliti, 1998 (49)	Italy	1,503	Retrospective cohort study	Methadone, oral naltrexone	12 months	Older age was associated with longer retention and was the only demographic variable associated with retention.
Dumchev, 2017 (44)	Ukraine	2,916	Secondary analysis of clinical data	Methadone, buprenorphine	12 months	Age was not significantly associated with retention.
Dupouy, 2013 (68)	France	1,507	Retrospective cohort study	Methadone, buprenorphine	30 months	Older age was associated with longer retention.
Fishman, 2010 (69)	United States	16	Retrospective chart review	Injectable naltrexone only	4 months	All participants in study were adolescents and young adults. Factors associated with adherence or retention were not explored in study.
Gonzalez, 2015 (59)	United States	80	RCT	Buprenorphine only	13 weeks	All participants in the study were young adults. No differences in retention between group treated with memantine vs. group treated with buprenorphine/naloxone.
Grella, 1997 (70)	United States	462	Experimental study	Methadone only	24 months	Older age was associated with longer retention.
Gryczynski, 2012 (38)	United States	181	Prospective cohort study	Methadone only	12 months	Age was not significantly associated with retention.
Guillou Landreat, 2014 (46)	France	67	Prospective cohort study	Buprenorphine only	10 years	Older age was associated with higher adherence.
Haddad, 2013 (48)	United States	266	Retrospective cohort study	Buprenorphine only	6 months ^a ; 12 months ^a	Older age was associated with longer retention at 6 months, but not at 12 months.
Hadland, 2018 (20)	United States	4,837	Retrospective cohort study	Methadone, buprenorphine, oral or injectable naltrexone	1 year	All participants in the study were adolescents or young adults. Timely receipt of treatment was associated with longer retention; retention was longer among group receiving methadone vs. groups receiving buprenorphine or naloxone.
Hser, 2014 (71)	United States	1,267	RCT	Methadone, buprenorphine	24 weeks	Older age was associated with longer retention.
Joe, 1975 (72)	United States	4,981	Secondary analysis of program data	Methadone only	12 months	Older age was associated with longer retention in methadone maintenance, adaptive (indefinite methadone maintenance).

Table continues

Table 1. Continued

First Author, Year (Reference No.)	Country	No. of Participants	Study Design	Type of MOUD	Max. Retention Time Frame	Age-Related Findings
Kellogg, 2006 (53)	United States	147	Retrospective cohort study	Methadone only	12 months	All patients in the study were adolescents or young adults. Age and consistent heroin use were associated with shorter retention; gender and race were unrelated to retention.
Kornor, 2006 (56)	Norway	75	Experimental study	Buprenorphine only	9 months	All patients in the study were young adults. Having fewer previous treatment episodes was related to longer retention.
Kumar, 2016 (39)	United States	113	Retrospective chart review	Buprenorphine only	90 days	Age was not significantly associated with retention.
Lambdin, 2014 (73)	Tanzania	629	Retrospective cohort study	Methadone only	12 months	Older age was associated with longer retention.
Lee, 2017 (74)	United States	139	Secondary analysis of RCT	Buprenorphine only	6 months	Older age was associated with longer retention.
Lo-Ciganic, 2019 (75)	United States	2,361	Retrospective cohort study	Buprenorphine only	52 weeks	Older age was associated with longer retention.
Marcovitz, 2016 (24)	United States	202	Retrospective chart review	Buprenorphine only	3 months	Older age was associated with longer retention.
Matson, 2013 (76)	United States	185	Retrospective chart review	Buprenorphine only	≥5 months	All participants in the study were adolescents. Factors associated with adherence or retention were not explored in this study.
Matson, 2014 (54)	United States	103	Retrospective cohort study	Buprenorphine only	1 year	All participants in the study were adolescents or young adults. Female sex, negative urine drug screen for opioids or for marijuana, positive urine drug screen for buprenorphine/naloxone were all associated with longer retention.
Mutlu, 2016 (50)	Turkey	112	Retrospective cohort study	Buprenorphine only	1 year	All participants in the study were adolescents. Buprenorphine/naloxone dose, length of inpatient treatment, duration of program retention, and comorbid psychiatric disorder were associated with longer retention; age of admission was not associated with retention; elevated liver enzyme levels were associated with better short-term retention but worse long-term retention.
Perreault, 2015 (40)	Canada	106	Prospective cohort study	Methadone only	12 months	Age was not significantly associated with retention.

Table continues

Table 1. Continued

First Author, Year (Reference No.)	Country	No. of Participants	Study Design	Type of MOUD	Max. Retention Time Frame	Age-Related Findings
Romero-Gonzalez, 2017 (77)	United States	63	Secondary analysis of RCT	Buprenorphine only	8 weeks	All participants in the study were young adults. No difference in retention between heroin users and prescription opioid users.
Ruisenor-Escudero, 2015 (78)	Afghanistan	83	Cross-sectional study	Methadone only	18 months	Older age was associated with longer retention.
Samples, 2018 (79)	United States	17,329	Retrospective cohort study	Buprenorphine only	180 days	Older age was associated with longer retention.
Saxon, 1996 (80)	United States	353	RCT	Methadone only	24 weeks	Older age was associated with longer retention.
Schiff, 2007 (81)	Israel	2,683	Retrospective chart review	Methadone only	13 months	Older age was associated with longer retention. The interaction of age and sex was not statistically significant.
Schuman-Olivier, 2014 (52)	United States	294	Retrospective chart review	Buprenorphine only	3 months ^a ; 12 months ^a	All participants in the study were adolescents or young adults. Older age was associated with longer retention.
Smyth, 2012 (82)	Ireland	100	Retrospective cohort study	Methadone, buprenorphine	12 months	All participants in the study were adolescents. Male participants more likely to exit treatment because of incarceration.
Smyth, 2018 (58)	Ireland	120	Retrospective cohort study	Methadone, buprenorphine	12 months	All participants in the study were adolescents. Having children, coming from a single-parent family, not being in an intimate relationship with another heroin user, and evidence of cocaine use just before treatment entry were associated with shorter retention; no difference among those treated with methadone vs. buprenorphine.
Soeffing, 2009 (45)	United States	255	Retrospective cohort study	Buprenorphine only	12 months	Age was not significantly associated with retention.
Vo, 2016 (51)	United States	56	Retrospective chart review	Buprenorphine, injectable naltrexone	24 weeks	All participants in the study were adolescents or young adults. No difference in retention of those <24 years old and those ≥24 years old or between those treated with buprenorphine vs. naltrexone; male participants were retained in treatment longer.

Table continues

Table 1. Continued

First Author, Year (Reference No.)	Country	No. of Participants	Study Design	Type of MOUD	Max. Retention Time Frame	Age-Related Findings
Vo, 2018 (83)	United States	14	Retrospective chart review	Injectable naltrexone only	16 weeks	All participants in the study were adolescents or young adults. Better adherence and retention were both associated with home-based delivery vs. clinic-based, but no statistical analysis done due to small sample size.
Warden, 2012 (55)	United States	152	RCT	Buprenorphine only	12 weeks	All participants in the study were adolescents or young adults. Adherence, early opioid-negative urine tests, use of any medications in the month prior to treatment entry, and lifetime nonheroin opioid use were associated with longer retention; prior 30-day hallucinogen use was associated with shorter retention
Weinstein, 2017 (23)	United States	1,237	Retrospective cohort study	Buprenorphine only	≥ 1 year ^a ; ≥ 2 years ^a	Older age was associated with longer retention.
Wilder, 2017 (41)	United States	189	Retrospective cohort study	Methadone only	60 days	Age was not significantly associated with retention

Abbreviations: Max., maximum; MOUD, medications for opioid use disorder; RCT, randomized controlled trial.

^a The time frame for analysis of the relationship between age and retention when multiple time frames are given for the length of follow-up. In cases where multiple time frames are included but none is highlighted with an asterisk, then survival analysis (usually Cox proportional hazards regression analysis) was conducted using the length of time from treatment initiation to dropout for each study participant.

and retention in this subgroup analysis ($n = 32$) because 6 of these studies included more than 1 MOUD type, though D'Ippoliti et al. (49) only explored the relationship between age and retention for those patients treated with methadone and not for those receiving naltrexone.

Of the 20 studies in which the authors looked at the relationship between age and retention for individuals treated with methadone, 75.0% reported a positive association, with

older age corresponding to a greater likelihood of retention in MOUD, though 1 study found a negative association (5.0%). Of the 15 studies exploring this relationship among individuals on buprenorphine, two-thirds (66.7%) found a positive association between age and retention. Finally, neither of the studies examining the relationship between age and retention in individuals treated with naltrexone found a significant association.

Table 2. Quality Assessment of Articles Exploring the Relationship Between Age and Retention in MOUD

Article Results	No. Of Studies	No. of RCTs	Risk Of Bias	Inconsistency	Indirectness	Imprecision	Publication Bias	Grade
Exploring relationship of age with retention	35	3	Low	Moderate	Not an issue	Some	Undetected	Low
Age was positively associated with retention	20	3	Low	None	Not an issue	Not an issue	Undetected	Low
Age was not associated with retention	11	0	Low	None	Not an issue	Very serious	Undetected	Very low

Abbreviation: RCT, randomized controlled trial.

Relationship between age and MOUD retention by measurement of retention

The relationship between age and retention in MOUD was explored relative to the length of retention measured in the study to see if studies measuring retention over longer times found different results related to the association between age and retention. Of the 32 studies that did not present mixed results, one-third (34.4%; $n = 11$) measured retention using a time frame shorter than 12 months and one-quarter (25.0%; $n = 8$) measured retention using a time frame longer than 12 months; the remaining studies (40.6%; $n = 13$) measured retention using a time frame of exactly 12 months or its equivalent (e.g., 52 weeks).

Similar percentages of studies that measured retention using a time frame shorter than 12 months or of exactly 12 months found a positive association between age and MOUD retention (63.6% and 61.5%, respectively); the remaining studies found no significant association between age and retention in MOUD. A larger percentage of studies that measured retention using a time frame longer than 12 months found a positive association between age and MOUD retention (75.0%); the only study to find a negative association between age and MOUD retention also used a time frame longer than 12 months (35).

Relationship between age and retention by treatment of age

The relationship between age and retention was further explored to see if this relationship varied on the basis of whether age was treated as a continuous or categorical variable. Of the 28 studies included in this analysis, 16 (57.1%) treated age as a continuous variable whereas the remainder treated age categorically. Four studies were excluded because it was not clear from the authors' discussion of the analysis how age was treated; all but 1 of these studies found no significant association between age and retention.

Among those studies in which age was treated continuously, nearly two-thirds (62.5%) found a positive association between age and retention in MOUD. A larger percentage of studies that treated age categorically (83.3%) found a positive association between age and MOUD retention. The 1 study that found a negative association between age and MOUD retention also treated age categorically.

Qualitative coding and analysis

Qualitative coding and analysis were conducted with a subset of 17 studies that had data segregated by age for adolescents and young adults, either as defined by the author or based on the age range of 10–24 years. All these studies explored retention in MOUD among adolescents and young adults, so the coding focused on factors related to retention (to the exclusion of adherence). After coding, the findings from these studies were thematically organized using Bronfenbrenner's Social Ecological Model (30). A total of 16 codes were used, which were grouped into themes according to 3 levels: individual, interpersonal, or institutional; no factors at the community or societal levels

were observed. Individual-level themes included age, adherence to MOUD (as a predictor of retention), demographics other than age (i.e., gender, race/ethnicity), drug use (past and current), physical health, psychiatric comorbidity, and treatment history for substance use disorders. Interpersonal themes encompassed relationships with others, including family members and intimate partners. Institutional themes focused on aspects of the MOUD program itself and included treatment modality, dose, structure, and timeliness (Figure 2).

At the individual level, age was not consistently associated with retention when looking only at adolescents and young adults. Two studies found age was unrelated to retention (50, 51), 1 reported younger age and retention were negatively related (52), and another study found younger age and retention were positively related (53). These findings contrast with the earlier trend seen in studies with samples that included older adults, which consistently observed that age was positively related to retention. This may be due to greater homogeneity within the adolescent and young-adult age group or a nonlinear relationship between age and retention, though it is important to note that Vo et al. (51) compared participants younger than 24 years with a separate sample of individuals aged 24 years or older and found no significant association between age and retention.

Study results were similarly inconsistent when exploring the associations among race, gender, and retention. One study found no association between either race or gender and retention (53) and 2 other studies (51, 54) presented opposite results with respect to the association between gender and retention. It is possible that these results are attributable to differences in the age ranges of participants, with participants in the study by Vo et al. (51) older than those in the other 2 studies; however, the type of medication used might have played more of a role in the difference in the reported results.

The research showed a consistent relationship between drug use and retention. In 3 studies (53–55), authors found that opioid use during treatment was negatively associated with retention; 1 of these studies (55) also found that hallucinogen use during treatment was negatively associated with retention. Two studies (54, 55) found adherence to MOUD was positively related to retention. The research exploring the associations of physical health issues (50, 55) and treatment history (50, 56) with retention demonstrated mixed results, with 1 study finding these variables positively associated with retention and a second study finding a negative association between the 2. Finally, in 1 study (50), psychiatric comorbidity was positively associated with retention.

At the interpersonal level, stressful family situations, such as drug use among immediate family members (57), being raised in a single-parent household, or having a child (58), were all negatively associated with retention. On the other hand, being in an intimate relationship with someone with a history of heroin use was positively associated with retention (58).

At the institutional level, 3 studies found the type of medication used was unrelated to retention (51, 58, 59); however, this finding is in contrast to the results of 2 studies

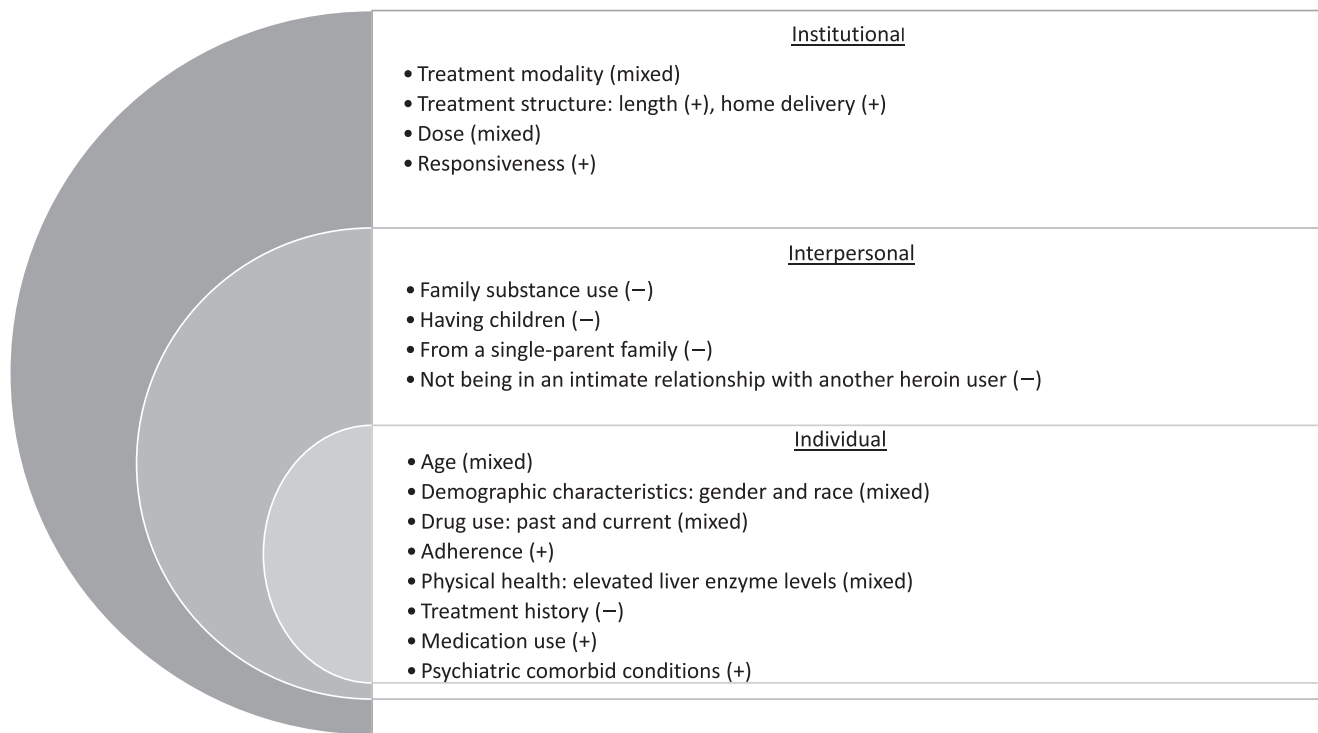


Figure 2. Factors associated with medications for opioid use disorder (MOUD) retention among young persons by levels of the social ecological model. +, consistent positive association between study variable and retention in MOUD; –, consistent negative association between study variable and retention in MOUD; mixed, inconsistent reports of the association between study variable and retention in MOUD.

(20, 60) in which greater retention was reported among individuals in methadone-based treatment compared with individuals in buprenorphine-based treatment. Results were not consistent when looking at the relationship between treatment dose and retention. One study (50) found a higher dose of buprenorphine was associated with longer retention, whereas another (57) found it was negatively associated with retention. Finally, timely provision of treatment was also positively associated with retention (20).

DISCUSSION

This systematic review was conducted to assess the state of the literature on the factors associated with adherence and retention to MOUD among adolescents and young adults experiencing OUD, a population often identified as at heightened risk for dropout from such treatment. Many studies examining adherence or retention in MOUD that included adolescents and young adults did not account for the possible association with age. Of the 631 articles that were excluded after full-text review, one-third (33.1%) did not focus explicitly on adherence or retention among young persons, nor did they account for associations with age in their analysis while including adolescents or young adults in the total sample. Many of these studies represented age categorically in analyses, using age groups that were too broad for meaningful analysis of the relationship between adolescent or young adult status and retention in MOUD

and too inconsistent for comparison of associations across studies. In other studies, the possible association with age was controlled for using statistical comparison of mean ages across groups. This approach, however, does not fully control for the possible association between age and retention, especially if the age range is skewed. Only in the study by Guillou Landreat et al. (46) did researchers look at adherence as an outcome.

In this review, we found that adolescents and young adults with OUD demonstrated heightened risk for dropout from MOUD in more than half of included articles. Although one-third of included studies did not find an association between younger age and increased risk of dropout from MOUD, the overall quality of these studies was lower. The studies in this group with larger sample sizes tended to be conducted in LMIC (37, 42, 44). This may point to the presence of potentially confounding contextual or structural factors specific to LMIC settings, such as cultural differences in attitudes toward substance use in younger age groups or different levels of acceptability or policy standards with respect to the use of MOUD among adolescents and/or young adults. Differences in the operationalization of retention in MOUD may also play a role in this discrepancy; the study by Armstrong et al. (42), for example, looked at departure from MOUD due specifically to relapse as an outcome rather than retention or attrition from MOUD more generally. It is also possible that these studies indicate there are other confounding factors responsible for

the observed relationship between older age and increasing retention; for example, del Rio et al. (43) found no association between age and retention after controlling for duration of opioid use. Finally, Perreault et al. (40), who did not find a significant association between age and retention, hypothesized that this may be due to “differences in sampling.”

Only 2 of the included studies found that adolescents and young adults had longer retention in MOUD than older adults. Of these, the study by Chaudhry et al. (34), although initially finding an association between younger age and longer retention, determined that this association was due to the presence of confounding variables. The authors corrected for this in multivariate analysis and their final results were consistent with those of most of the other studies included in our sample. In the other study, by Cousins et al. (35), important confounders, such as duration of drug use history, may not have been controlled for.

Stratifying by MOUD type (i.e., methadone, buprenorphine, and naltrexone) revealed that the evidence for a positive association between age and MOUD retention is fairly consistent for both methadone and buprenorphine, with a majority of studies finding such a relationship for both medications (75.0% and 66.7%, respectively). Neither of the studies examining the relationship between age and retention for naltrexone found a significant association, which points to the need for more studies examining retention in naltrexone maintenance to explore the association with age and to include adolescents and young adults in the sample.

Exploring the relationship between age and retention in MOUD stratified by how retention was measured yielded similar results. Most of the studies that measured retention using a time frame shorter than 12 months or exactly 12 months found a positive association between age and retention in MOUD (63.6% and 61.5%, respectively). A larger percentage of studies that measured retention using a time frame longer than 12 months found a positive association between age and retention in MOUD (75.0%); however, only 8 studies (25%) included in the subgroup analysis used a time frame longer than 12 months. This highlights the need for more studies examining retention in MOUD beyond 12 months among adolescents and young adults.

Similar results were observed when looking at if the relationship between age and retention varied by whether age was treated continuously and categorically. Although 62.5% of studies that treated age continuously in analysis found a positive association between age and retention, a higher percentage (83.3%) of those that treated age categorically in analysis found a positive association. These findings point to the possibility of a nonlinear relationship between age and MOUD retention.

Few studies ($n = 17$; 6%) exploring retention in MOUD as an outcome focused on adolescents or young adults specifically, allowing for an analysis of the facilitators of and barriers to adherence and retention in MOUD in this population. The results of the qualitative analysis of this group of articles demonstrated that the relationship between age and retention did not stay constant when assessing the

association between age and retention within adolescents and young adults, suggesting some level of homogeneity within this age group or a nonlinear relationship between age and retention. Qualitative analysis revealed a number of factors at different levels that are associated with retention in MOUD among adolescents and young adults, but it is also important to note the factors that were absent from analysis and were not seen in any of the included research studies, such as individual behaviors, socioeconomic status, the role of social networks, or structural factors related to MOUD accessibility and availability. Additional research exploring the possible role of these factors in adherence to and retention in MOUD among adolescents and young adults would help inform strategies to better engage this group in treatment.

The most consistently observed association in this review was that younger age is linked to lower rates of retention. This association was also seen in other systematic reviews on the topic of retention in MOUD (28). Of the final included articles in the Brorson et al. (28) 2013 systematic review of risk factors for dropout from MOUD, younger age was the most frequently cited risk factor for MOUD dropout. Unlike in the present review's final sample, Brorson et al. (28) did not find any studies that contradicted this relationship; indeed, of all risk factors, younger age was the only factor that consistently predicted dropout. Similarly, the observation from qualitative synthesis that youth receiving methadone treatment seem to experience better retention than youth receiving buprenorphine or naltrexone is consistent with the findings in the reviews by Hopfer et al. (26), Minozzi et al. (25), and Timko et al. (27). Our review adds to the findings of these previous reviews by exploring the factors, especially at the interpersonal and institutional levels, that are possibly responsible for the shorter retention in MOUD seen among adolescents and young adults. Although it is not possible to draw any definitive statements from the quantitative synthesis in this review, our findings do point to factors that merit more research. Future research, especially qualitative and mixed methods research, should explore possible factors that explain the relationship between younger age and increased dropout from MOUD. Higher-quality research studies exploring adherence to and retention in MOUD and that stratify populations by age groups to compare adolescents and young adults with older adults would also add substantially to the literature. These findings also point to the need for standardization in the approach to age-based analyses in quantitative research on MOUD adherence and retention where the sample includes adolescents and young adults. Finally, these findings indicate that providers of MOUD should consider tailored approaches to retain adolescents and young adults in treatment. At the individual and interpersonal levels, this would include screening and referral services for factors that might compromise retention, such as concurrent drug use, physical health, psychiatric comorbidity, or family stress and conflict. At the institutional level, MOUD providers should ensure that adolescents and young adults receive adequate dosing of medication and should adjust policies and procedures so younger program participants have opportunities for more flexible dosing and pick-up schedules.

LIMITATIONS

This review faces the same limitation addressed by Brorson et al. (28) in that differences in the operationalization of retention and in reason for dropout may act as a confounder of our observations. LMIC are underrepresented in our final sample of articles, whereas the United States is overrepresented. Studies conducted in LMIC that included adolescents and young adults and would otherwise meet our inclusion criteria did not segregate their data by age group, nor did they assess the associations between age and adherence or retention in MOUD; instead, these studies tended to focus on clinical process improvement (e.g., the study by Hasečić et al. (61)), demonstrating different research priorities compared with studies done in high-income countries. Few qualitative studies explore adherence and retention in MOUD in this age group. As such, the existing body of research shows an association between age and adherence and retention without exploring the specific factors responsible for lower rates of adherence and retention among adolescents and young adults.

The quality of the articles included for quantitative analysis and synthesis in this review was, overall, low, according to Grading of Recommendations Assessment, Development, and Evaluation standards. More high-level research studies, especially randomized controlled trials, are needed to boost the level of evidence of the association between age and retention in MOUD.

A total of 22 articles were excluded from this review because they were in languages not spoken by any of the reviewers (specifically, Chinese, French, German, Norwegian, Persian, and Swedish). Given the international focus of this review, we worked to ensure we could review articles in as many languages as possible, but we were unable to cover the languages of all potentially eligible articles. In addition, Embase introduced indexing for adolescents late and subsequent indexing in that database is not as sensitive as the Medline indexing for this concept, so it is possible that some studies in which adolescents were part of the sample were overlooked in this review because of this limitation.

CONCLUSION

This review upholds the association seen in previous reviews that adolescents and young adults experience shorter retention in MOUD than individuals in older age groups. Studies that explore retention in MOUD as an outcome and that include adolescents or young adults in their sample should adequately stratify their findings or otherwise account for the possible difference in retention in this age group. Much of the challenge in drawing definitive conclusions in this review was due to the lack of a standardized definition of concepts such as “adolescent,” “young adult,” “adherence,” and “retention”; as such, the research community should develop and use standardized definitions for different age groups of interest as well as for the concepts of adherence and retention, to facilitate comparison across large groups of studies. More studies are needed, especially randomized controlled trials and qualitative and mixed methods studies, that explicitly look at what mechanisms are

at play that lead adolescents and young adults to drop out of MOUD earlier. Individual, interpersonal, and institutional factors, such as concurrent drug use, family conflict, and MOUD dosage and flexibility, play key roles in moderating or mediating this relationship and suggest possible avenues for better tailoring MOUD to meet the needs of this age group. It is clear that adolescents and young adults experience unique needs when it comes to continued engagement in MOUD, and this review highlights the main factors that should be addressed.

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REFERENCES

1. United Nations Office on Drugs and Crime. *World Drug Report: 2018*. 2018. Available from: https://www.unodc.org/wdr2018/prelaunch/WDR18_Booklet_1_EXSUM.pdf. Accessed March 16, 2020.
2. Degenhardt L, Charlson F, Ferrari A, et al. The global burden of disease attributable to alcohol and drug use in 195 countries and territories, 1990-2016: a systematic analysis for the global burden of disease study 2016. *Lancet Psychiatry*. 2018;5(12):987–1012.

3. Veilleux JC, Colvin PJ, Anderson J, et al. A review of opioid dependence treatment: pharmacological and psychosocial interventions to treat opioid addiction. *Clin Psychol Rev.* 2010;30(2):155–166.
4. Degenhardt L, Stockings E, Patton G, et al. The increasing global health priority of substance use in young people. *Lancet Psychiatry.* 2016;3(3):251–264.
5. Chambers RA, Taylor JR, Potenza MN. Developmental neurocircuitry of motivation in adolescence: a critical period of addiction vulnerability. *Am J Psychiatry.* 2003;160(6):1041–1052.
6. Connery HS. Medication-assisted treatment of opioid use disorder: review of the evidence and future directions. *Harv Rev Psychiatry.* 2015;23(2):63–75.
7. American Society for Addiction Medicine. *National Practice Guideline for the Use of Medications in the Treatment of Addiction Involving Opioid Use.* 2015. Available from: <https://www.asam.org/docs/default-source/practice-support/guidelines-and-consensus-docs/asam-national-practice-guideline-supplement.pdf>. Accessed March 16, 2020.
8. Lawrinson P, Ali R, Buavirat A, et al. Key findings from the WHO collaborative study on substitution therapy for opioid dependence and HIV/AIDS. *Addiction.* 2008;103(9):1484–1492.
9. Nielsen S, Larance B, Degenhardt L, et al. Opioid agonist treatment for pharmaceutical opioid dependent people. *Cochrane Database Syst Rev.* 2016;(5):CD011117.
10. Subramaniam GA, Warden D, Minhajuddin A, et al. Predictors of abstinence: National Institute of drug abuse multisite buprenorphine/naloxone treatment trial in opioid-dependent youth. *J Am Acad Child Adolesc Psychiatry.* 2011;50(11):1120–1128.
11. Dupouy J, Palmaro A, Fatseas M, et al. Mortality associated with time in and out of buprenorphine treatment in French office-based general practice: a 7-year cohort study. *Ann Fam Med.* 2017;15(4):355–358.
12. Ledberg A. Mortality related to methadone maintenance treatment in Stockholm, Sweden, during 2006–2013. *J Subst Abuse Treat.* 2017;74:35–41.
13. Arefnasab Z, Rahimi C, Mohammadi N, et al. The effect of methadone maintenance treatment (MMT) on the mental health of opium and heroin addicts. *J Iran Psychol.* 2007;4(13):63–52.
14. Evans E, Li L, Min J, et al. Mortality among individuals accessing pharmacological treatment for opioid dependence in California, 2006–10. *Addiction.* 2015;110(6):996–1005.
15. Degenhardt L, Mathers BM, Wirtz AL, et al. What has been achieved in HIV prevention, treatment and care for people who inject drugs, 2010–2012? A review of the six highest burden countries. *Int J Drug Policy.* 2014;25(1):53–60.
16. Larney S, Peacock A, Leung J, et al. Global, regional, and country-level coverage of interventions to prevent and manage HIV and hepatitis C among people who inject drugs: a systematic review. *Lancet Glob Health.* 2017;5(12):e1208–e1220.
17. Chang DC, Klimas J, Wood E, et al. Medication-assisted treatment for youth with opioid use disorder: current dilemmas and remaining questions. *Am J Drug Alcohol Abuse.* 2018;44(2):143–146.
18. World Health Organization. *Guidelines for the Psychosocially Assisted Pharmacological Treatment of Opioid Dependence.* 2009. Available from: https://www.who.int/substance_abuse/publications/opioid_dependence_guidelines.pdf. Accessed March 16, 2020.
19. American Academy of Pediatrics Committee on Substance Use and Prevention. Medication-assisted treatment of adolescents with opioid use disorders. *Pediatrics.* 2016;138(3):e20161893.
20. Hadland SE, Bagley SM, Rodean J, et al. Receipt of timely addiction treatment and association of early medication treatment with retention in care among youths with opioid use disorder. *JAMA Pediatr.* 2018;172(11):1029–1037.
21. Feder KA, Krawczyk N, Saloner B. Medication-assisted treatment for adolescents in specialty treatment for opioid use disorder. *J Adolesc Health.* 2017;60(6):747–750.
22. Knopf A. Maintenance buprenorphine for adults, but tapers for teens: why? *Brown University Child Adolesc Behav Lett.* 2016;32(12):4–5.
23. Weinstein ZM, Kim HW, Cheng DM, et al. Long-term retention in office based opioid treatment with buprenorphine. *J Subst Abuse Treat.* 2017;74:65–70.
24. Marcovitz DE, McHugh RK, Volpe J, et al. Predictors of early dropout in outpatient buprenorphine/naloxone treatment. *Am J Addict.* 2016;25(6):472–477.
25. Minozzi S, Amato L, Bellisario C, et al. Maintenance treatments for opiate-dependent adolescents. *Cochrane Database Syst Rev.* 2014;(6):CD007210.
26. Hopfer CJ, Khuri E, Crowley TJ, et al. Adolescent heroin use: a review of the descriptive and treatment literature. *J Subst Abuse Treat.* 2002;23(3):231–237.
27. Timko C, Schultz NR, Cucciare MA, et al. Retention in medication-assisted treatment for opiate dependence: a systematic review. *J Addict Dis.* 2016;35(1):22–35.
28. Brorson HH, Ajo Arnevik E, Rand-Hendriksen K, et al. Drop-out from addiction treatment: a systematic review of risk factors. *Clin Psychol Rev.* 2013;33(8):1010–1024.
29. Viera A, Bromberg DJ, Whittaker S, et al. *Medication-assisted treatment adherence and retention among adolescents and young adults with opioid use disorders.* 2018. Available from: https://www.crd.york.ac.uk/prospetro/display_record.php?ID=CRD42018117074. Accessed March 16, 2020.
30. Bronfenbrenner U. Toward an experimental ecology of human development. *Am Psychol.* 1977;32(7):513–531.
31. Guyatt G, Oxman AD, Akl EA, et al. GRADE guidelines: 1. Introduction—GRADE evidence profiles and summary of findings tables. *J Clin Epidemiol.* 2011;64(4):383–394.
32. Ryan R, Hill S. *How to GRADE the quality of the evidence.* 2016. <https://opal.latrobe.edu.au/ndownloader/files/12400511>. Accessed March 16, 2020.
33. Moher D, Liberati A, Tetzlaff J, et al. Preferred reporting items for systematic reviews and meta-analyses: the PRISMA statement. *PLoS Med.* 2009;6(7):e1000097.
34. Chaudhry ZA, Sultan J, Alam F. Predictors for retention in treatment with a UK community-based naltrexone programme for opioid dependence. *The Psychiatrist.* 2012;36(6):218–224.
35. Cousins G, Boland F, Barry J, et al. J-shaped relationship between supervised methadone consumption and retention in methadone maintenance treatment (MMT) in primary care: national cohort study. *Drug Alcohol Depend.* 2017;173:126–131.
36. Bartu A, Freeman NC, Gawthorne GS, et al. Characteristics, retention and readmissions of opioid-dependent clients treated with oral naltrexone. *Drug Alcohol Rev.* 2002;21(4):335–340.
37. Dhawan A, Chopra A. Does buprenorphine maintenance improve the quality of life of opioid users? *Indian J Med Res.* 2013;137(1):130–135.
38. Gryczynski J, Kinlock TW, Kelly SM, et al. Opioid agonist maintenance for probationers: patient-level predictors of

- treatment retention, drug use, and crime. *Subst Abus.* 2012; 33(1):30–39.
39. Kumar N, Stowe ZN, Han X, et al. Impact of early childhood trauma on retention and phase advancement in an outpatient buprenorphine treatment program. *Am J Addict.* 2016;25(7): 542–548.
 40. Perreault M, Julien D, White ND, et al. Psychological predictors of retention in a low-threshold methadone maintenance treatment for opioid addicts: a 1-year follow-up study. *Subst Use Misuse.* 2015;50(1):24–31.
 41. Wilder CM, Hosta D, Winhusen T. Association of methadone dose with substance use and treatment retention in pregnant and postpartum women with opioid use disorder. *J Subst Abuse Treat.* 2017;80:33–36.
 42. Armstrong G, Kermode M, Sharma C, et al. Opioid substitution therapy in Manipur and Nagaland, north-East India: operational research in action. *Harm Reduct J.* 2010; 7:29.
 43. del Rio M, Mino A, Perneger TV. Predictors of patient retention in a newly established methadone maintenance treatment programme. *Addiction.* 1997;92(10):1353–1360.
 44. Dumchev K, Dvoryak S, Chernova O, et al. Retention in medication-assisted treatment programs in Ukraine-identifying factors contributing to a continuing HIV epidemic. *Int J Drug Policy.* 2017;48:44–53.
 45. Soeffing JM, Martin LD, Fingerhuth MI, et al. Buprenorphine maintenance treatment in a primary care setting: outcomes at 1 year. *J Subst Abuse Treat.* 2009;37(4): 426–430.
 46. Guillo Landreat M, Seville-Rivain V, Victorri Vigneau C, et al. Buprenorphine prescription compliance: an original observational and longitudinal study. *J Psychoactive Drugs.* 2014;46(2):162–167.
 47. Babst DV, Chambers CD, Warner A. Patient characteristics associated with retention in a methadone maintenance program. *Br J Addict Alcohol Other Drugs.* 1971;66(3): 195–204.
 48. Haddad MS, Zelenev A, Altice FL. Integrating buprenorphine maintenance therapy into federally qualified health centers: real-world substance abuse treatment outcomes. *Drug Alcohol Depend.* 2013;131(1–2):127–135.
 49. D'Ippoliti D, Davoli M, Perucci CA, et al. Retention in treatment of heroin users in Italy: the role of treatment type and of methadone maintenance dosage. *Drug Alcohol Depend.* 1998;52(2):167–171.
 50. Mutlu C, Demirci AC, Yalcin O, et al. One-year follow-up of heroin-dependent adolescents treated with buprenorphine/naloxone for the first time in a substance treatment unit. *J Subst Abuse Treat.* 2016;67:1–8.
 51. Vo HT, Robbins E, Westwood M, et al. Relapse prevention medications in community treatment for young adults with opioid addiction. *Subst Abus.* 2016;37(3):392–397.
 52. Schuman-Olivier Z, Weiss RD, Hoepfner BB, et al. Emerging adult age status predicts poor buprenorphine treatment retention. *J Subst Abuse Treat.* 2014;47(3):202–212.
 53. Kellogg S, Melia D, Khuri E, et al. Adolescent and young adult heroin patients: drug use and success in methadone maintenance treatment. *J Addict Dis.* 2006;25(3):15–25.
 54. Matson SC, Hobson G, Abdel-Rasoul M, et al. A retrospective study of retention of opioid-dependent adolescents and young adults in an outpatient buprenorphine/naloxone clinic. *J Addict Med.* 2014;8(3): 176–182.
 55. Warden D, Subramaniam GA, Carmody T, et al. Predictors of attrition with buprenorphine/naloxone treatment in opioid dependent youth. *Addict Behav.* 2012;37(9):1046–1053.
 56. Kornør H, Waal H, Ali RL. Abstinence-orientated buprenorphine replacement therapy for young adults in out-patient counselling. *Drug Alcohol Rev.* 2006;25(2): 123–130.
 57. Dayal P, Balhara YPS. A naturalistic study of predictors of retention in treatment among emerging adults entering first buprenorphine maintenance treatment for opioid use disorders. *J Subst Abuse Treat.* 2017;80:1–5.
 58. Smyth BP, Elmusharaf K, Cullen W. Opioid substitution treatment and heroin dependent adolescents: reductions in heroin use and treatment retention over twelve months. *BMC Pediatr.* 2018;18(1):151.
 59. Gonzalez G, DiGirolamo G, Romero-Gonzalez M, et al. Memantine improves buprenorphine/naloxone treatment for opioid dependent young adults. *Drug Alcohol Depend.* 2015; 156(ebs, 7513587):243–253.
 60. Bell J, Mutch C. Treatment retention in adolescent patients treated with methadone or buprenorphine for opioid dependence: a file review. *Drug Alcohol Rev.* 2006;25(2): 167–171.
 61. Hasečić H, Mehić-Basara N, Pokrajac M, et al. Use of methadone therapy at the Institute for Alcoholism and Drug Addiction in the Canton of Sarajevo. *Med Arh.* 2003;57(5–6 Suppl 1):29–32.
 62. Anderson JF, Warren LD. Client retention in the British Columbia methadone program, 1996–1999. *Can J Public Health.* 2004;95(2):104–109.
 63. Burns L, Randall D, Hall WD, et al. Opioid agonist pharmacotherapy in New South Wales from 1985 to 2006: patient characteristics and patterns and predictors of treatment retention. *Addiction.* 2009;104(8):1363–1372.
 64. Burns L, Gisev N, Larney S, et al. A longitudinal comparison of retention in buprenorphine and methadone treatment for opioid dependence in New South Wales, Australia. *Addiction.* 2015;110(4):646–655.
 65. Condelli WS. Strategies for increasing retention in methadone programs. *J Psychoactive Drugs.* 1993;25(2):143–147.
 66. Damian AJ, Mendelson T, Agus D. Predictors of buprenorphine treatment success of opioid dependence in two Baltimore City grassroots recovery programs. *Addict Behav.* 2017;73:129–132.
 67. Deck D, Carlson MJ. Retention in publicly funded methadone maintenance treatment in two Western States. *J Behav Health Serv Res.* 2005;32(1):43–60.
 68. Dupouy J, Dassieu L, Bourrel R, et al. Effectiveness of drug tests in outpatients starting opioid substitution therapy. *J Subst Abuse Treat.* 2013;44(5):515–521.
 69. Fishman MJ, Winstanley EL, Curran E, et al. Treatment of opioid dependence in adolescents and young adults with extended release naltrexone: preliminary case-series and feasibility. *Addiction.* 2010;105(9):1669–1676.
 70. Grella CE, Wugalter SE, Anglin MD. Predictors of treatment retention in enhanced and standard methadone maintenance treatment for HIV risk reduction. *J Drug Issues.* 1997;27(2): 203–224.
 71. Hser YI, Saxon AJ, Huang D, et al. Treatment retention among patients randomized to buprenorphine/naloxone compared to methadone in a multi-site trial. *Addiction.* 2014; 109(1):79–87.
 72. Joe GW, Simpson DD. Retention in treatment of drug users: 1971–1972 DARP admissions. *Am J Drug Alcohol Abuse.* 1975;2(1):63–71.
 73. Lambdin BH, Masao F, Chang O, et al. Methadone treatment for HIV prevention-feasibility, retention, and predictors of attrition in Dar Es Salaam, Tanzania: a retrospective cohort study. *Clin Infect Dis.* 2014;59(5):735–742.

74. Lee CS, Liebschutz JM, Anderson BJ, et al. Hospitalized opioid-dependent patients: exploring predictors of buprenorphine treatment entry and retention after discharge. *Am J Addict.* 2017;26(7):667–672.
75. Lo-Ciganic WH, Donohue JM, Kim JY, et al. Adherence trajectories of buprenorphine therapy among pregnant women in a large state Medicaid program in the United States. *Pharmacoepidemiol Drug Saf.* 2019;28(1):80–89.
76. Matson SC, Hobson G, Abdel-Rasoul M, et al. Opiate abstinence among adolescents engaged in an outpatient buprenorphine clinic for the treatment of opiate dependence. *J Adolesc Health.* 2013;52(2):S17–S.
77. Romero-Gonzalez M, Shahanaghi A, DiGirolamo GJ, et al. Buprenorphine-naloxone treatment responses differ between young adults with heroin and prescription opioid use disorders. *Am J Addict.* 2017;26(8):838–844.
78. Ruisenor-Escudero H, Vu A, Wirtz AL, et al. Cross-sectional assessments of participants' characteristics and loss to follow-up in the first opioid substitution therapy pilot program in Kabul. *Harm Reduct J.* 2015;12:28.
79. Samples H, Williams AR, Olfson M, et al. Risk factors for discontinuation of buprenorphine treatment for opioid use disorders in a multi-state sample of Medicaid enrollees. *J Subst Abuse Treat.* 2018;95:9–17.
80. Saxon AJ, Wells EA, Fleming C, et al. Pre-treatment characteristics, program philosophy and level of ancillary services as predictors of methadone maintenance treatment outcome. *Addiction.* 1996;91(8):1197–1209.
81. Schiff M, Levit S, Moreno RC, et al. Retention and illicit drug use among methadone patients in Israel: a gender comparison. *Addict Behav.* 2007;32(10):2108–2119.
82. Smyth BP, Fagan J, Kernan K. Outcome of heroin-dependent adolescents presenting for opiate substitution treatment. *J Subst Abuse Treat.* 2012;42(1):35–44.
83. Vo HT, Burgower R, Rozenberg I, et al. Home-based delivery of XR-NTX in youth with opioid addiction. *J Subst Abuse Treat.* 2018;85:84–89.